

Assignment No 26

Name : Shivam Ramakant Deshmane

Domain : AI & ML

College : T.B Girwalkar polytechnic Ambajogai

Q1.

```
*** First 10 Rows:
      Feature1  Feature2
0 -8.033836   6.831299
1 -8.767448  -6.918116
2  6.172741   1.407361
3 -2.783673   7.989826
4 -6.038777  -7.165762
5 -6.298392  -7.479368
6 -7.046269   7.256281
7 -6.690307  -6.915276
8  3.892545   1.873504
9 -7.317000  -6.861846
```

```
Shape of Dataset:
(500, 2)
```

Q2.

5

```
# Import StandardScaler
from sklearn.preprocessing import StandardScaler
import pandas as pd

# Create StandardScaler object
scaler = StandardScaler()

# Apply Standard Scaling
X_scaled = scaler.fit_transform(df)

# Display first 5 rows of scaled data
scaled_df = pd.DataFrame(X_scaled, columns=["Feature1", "Feature2"])

print("First 5 Rows of Scaled Data:")
print(scaled_df.head())
```

First 5 Rows of Scaled Data:

	Feature1	Feature2
0	-0.888474	0.629907
1	-1.028908	-1.562875
2	1.831069	-0.235113
3	0.116557	0.814671
4	-0.506563	-1.602370

Q3.

```
▶ # Import KMeans
from sklearn.cluster import KMeans

# Create KMeans model
kmeans = KMeans(n_clusters=3, random_state=42)

# Fit the model and predict cluster labels
clusters = kmeans.fit_predict(X_scaled)

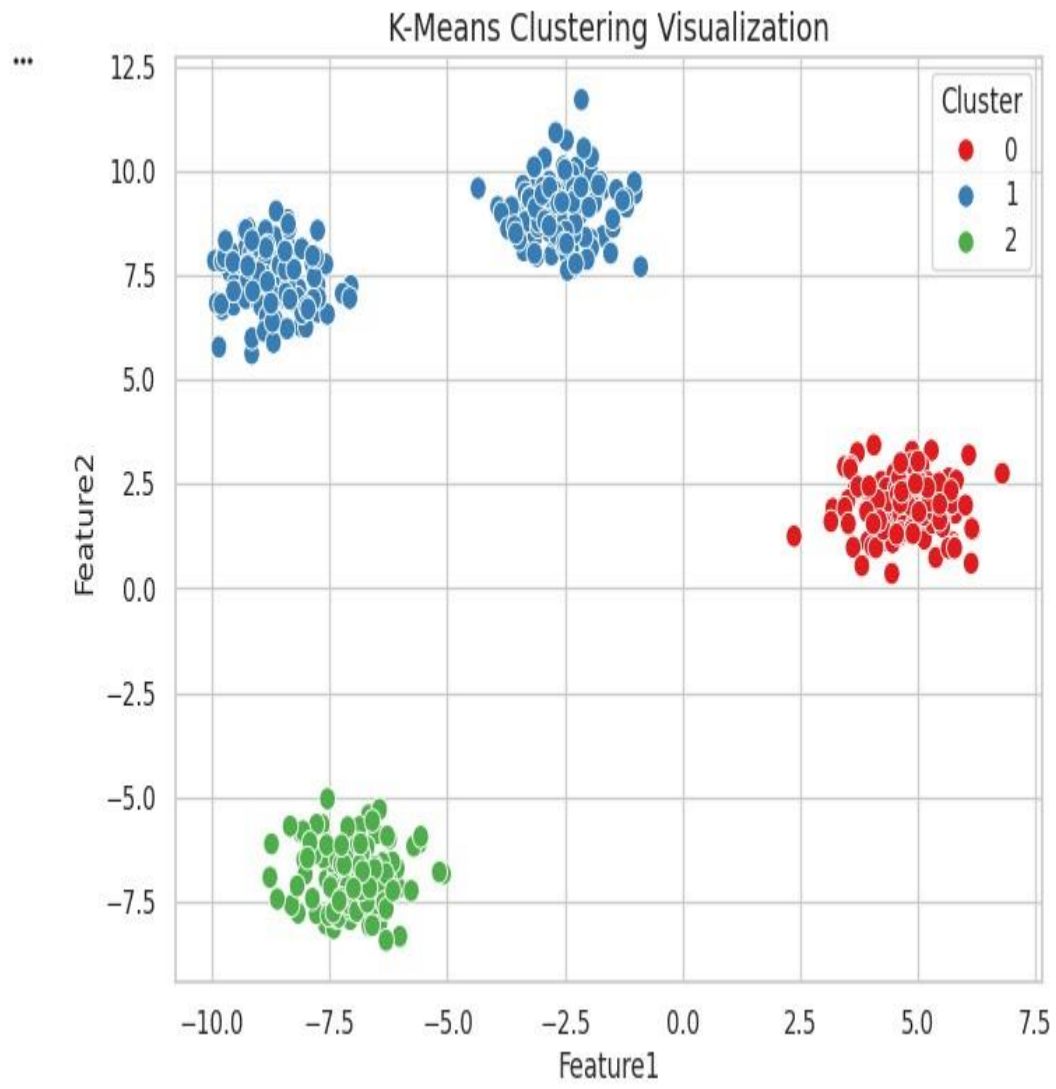
# Add cluster labels to the DataFrame
df["Cluster"] = clusters

# Display the first 10 rows
print("First 10 Rows of Updated DataFrame:")
print(df.head(10))
```

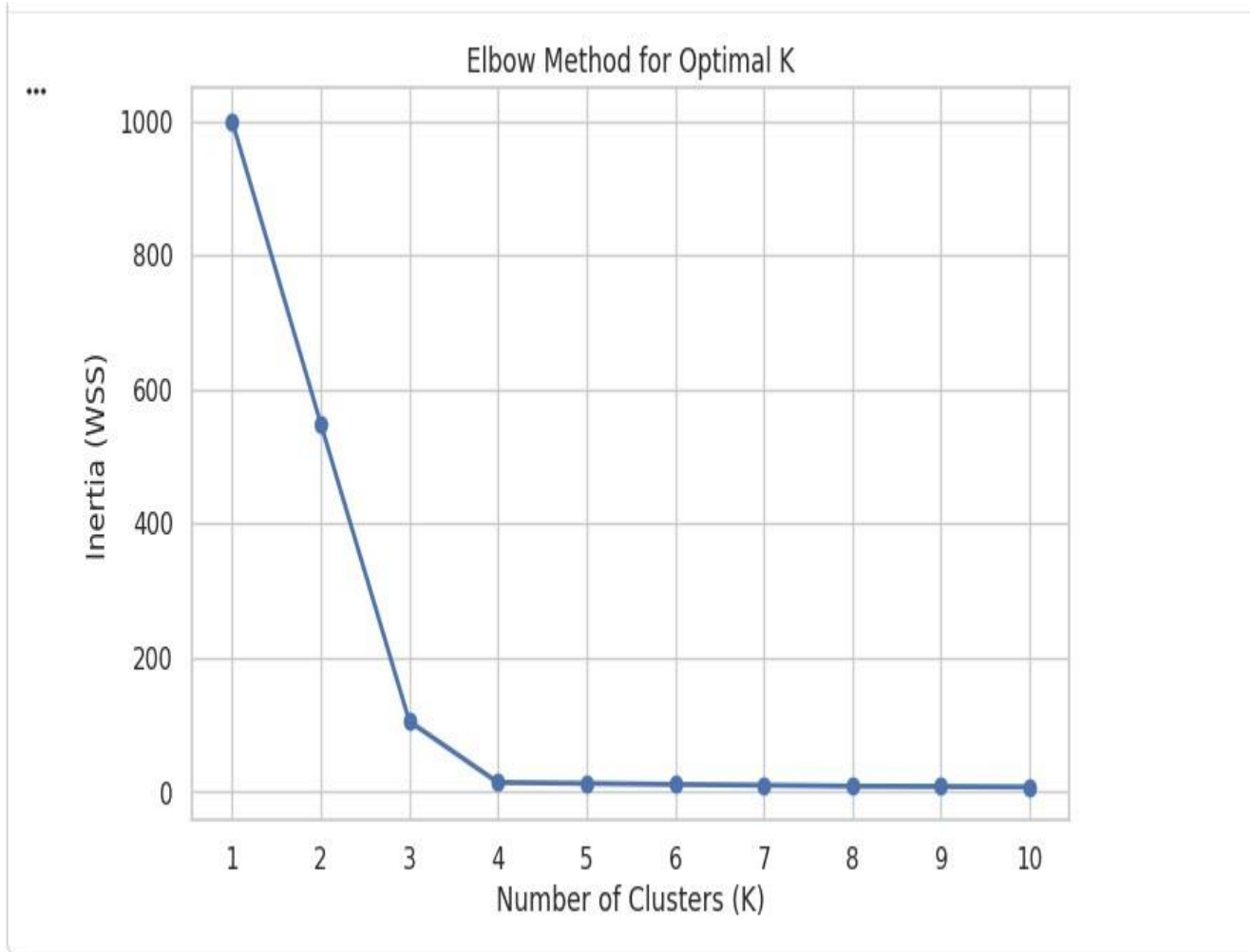
... First 10 Rows of Updated DataFrame:

	Feature1	Feature2	Cluster
0	-8.033836	6.831299	1
1	-8.767448	-6.918116	2
2	6.172741	1.407361	0
3	-2.783673	7.989826	1
4	-6.038777	-7.165762	2
5	-6.298392	-7.479368	2
6	-7.046269	7.256281	1
7	-6.690307	-6.915276	2
8	3.892545	1.873504	0
9	-7.317000	-6.861846	2

Q4.



Q5.



Q6.

*** Updated DataFrame:

	Feature1	Feature2	Cluster
0	-8.033836	6.831299	1
1	-8.767448	-6.918116	2
2	6.172741	1.407361	0
3	-2.783673	7.989826	3
4	-6.038777	-7.165762	2
5	-6.298392	-7.479368	2
6	-7.046269	7.256281	1
7	-6.690307	-6.915276	2
8	3.892545	1.873504	0
9	-7.317000	-6.861846	2

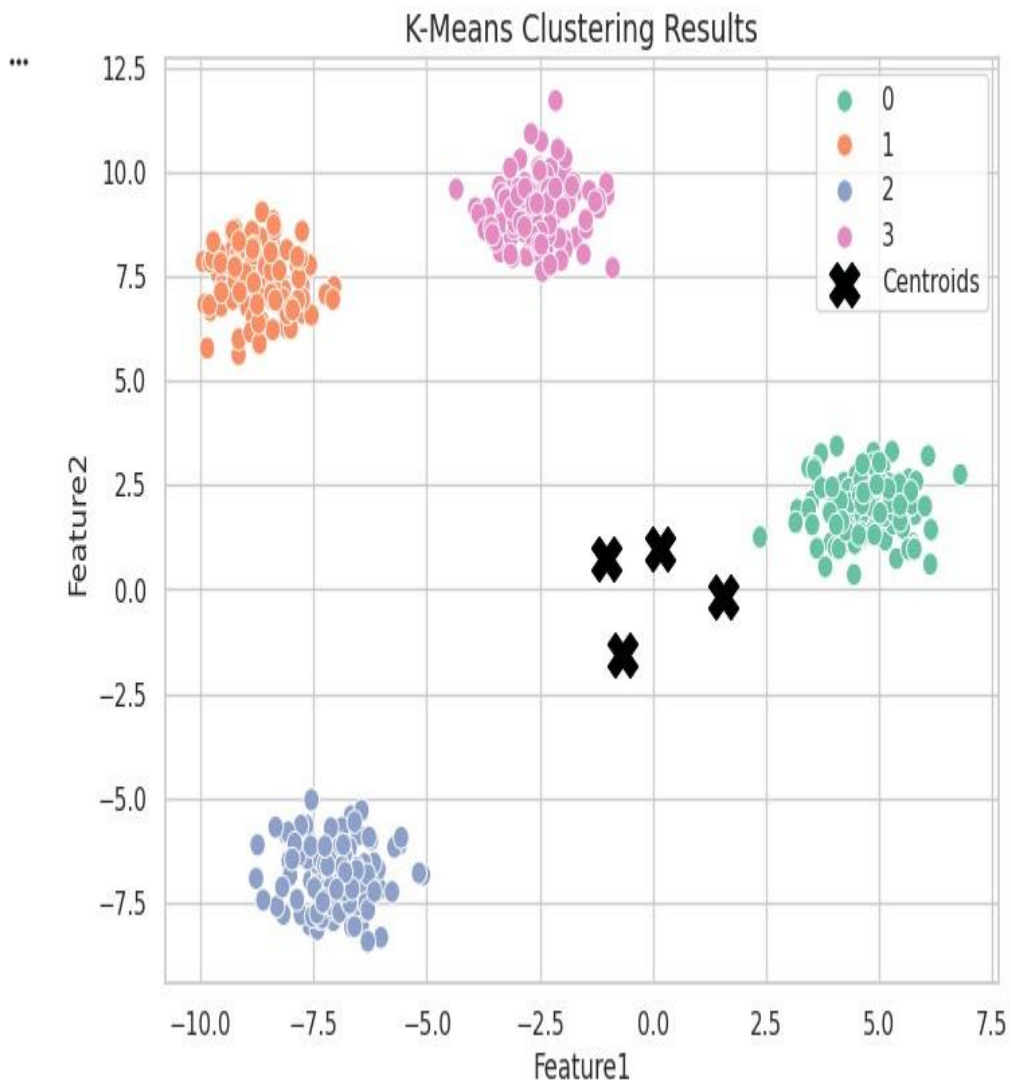
Number of Data Points in Each Cluster:

Cluster

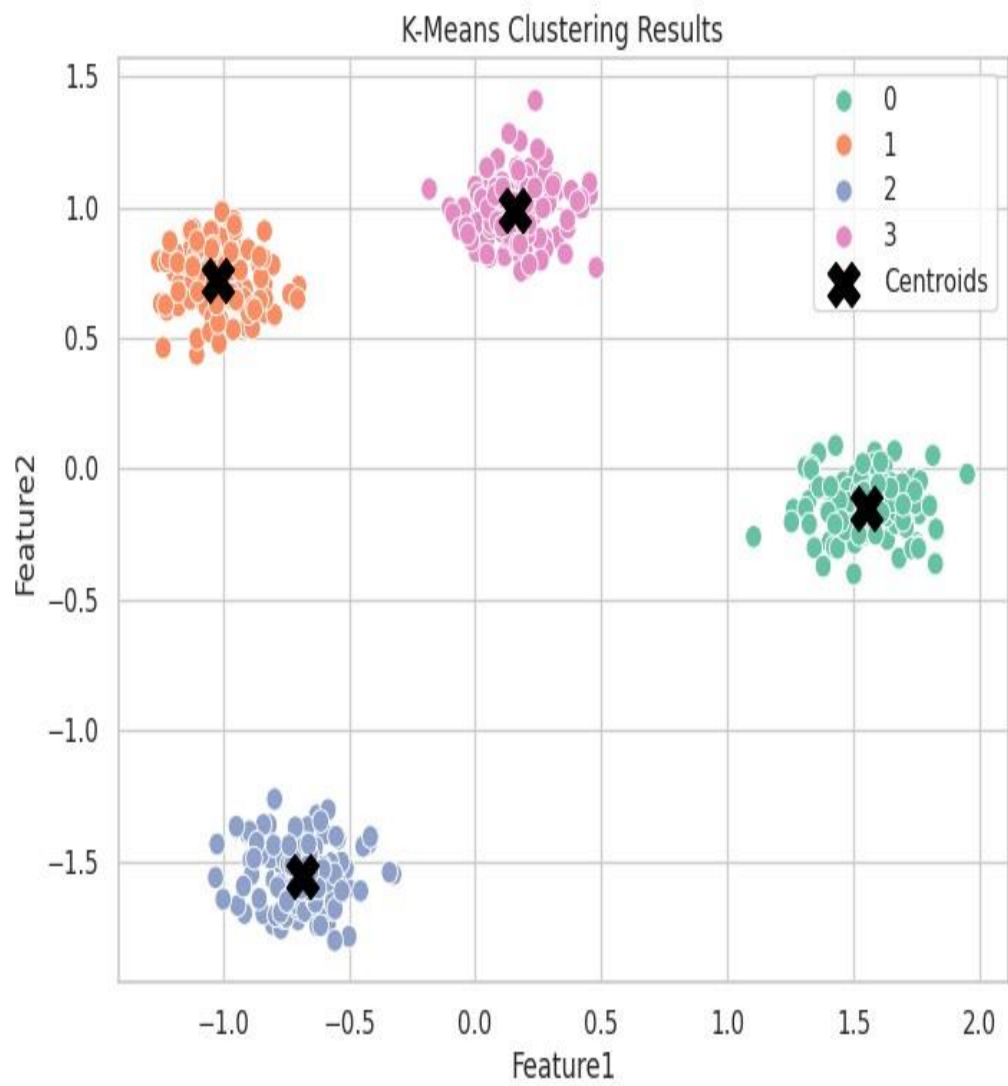
0	125
1	125
2	125
3	125

Name: count, dtype: int64

Q7.



...



Q8.

```
... First 5 Rows:
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

```
Data with Cluster Labels:
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	
5	5.4	3.9	1.7	0.4	
6	4.6	3.4	1.4	0.3	
7	5.0	3.4	1.5	0.2	
8	4.4	2.9	1.4	0.2	
9	4.9	3.1	1.5	0.1	

	Cluster
0	1
1	2
2	2
3	2
4	1
5	1
6	1
7	1
8	2
9	2

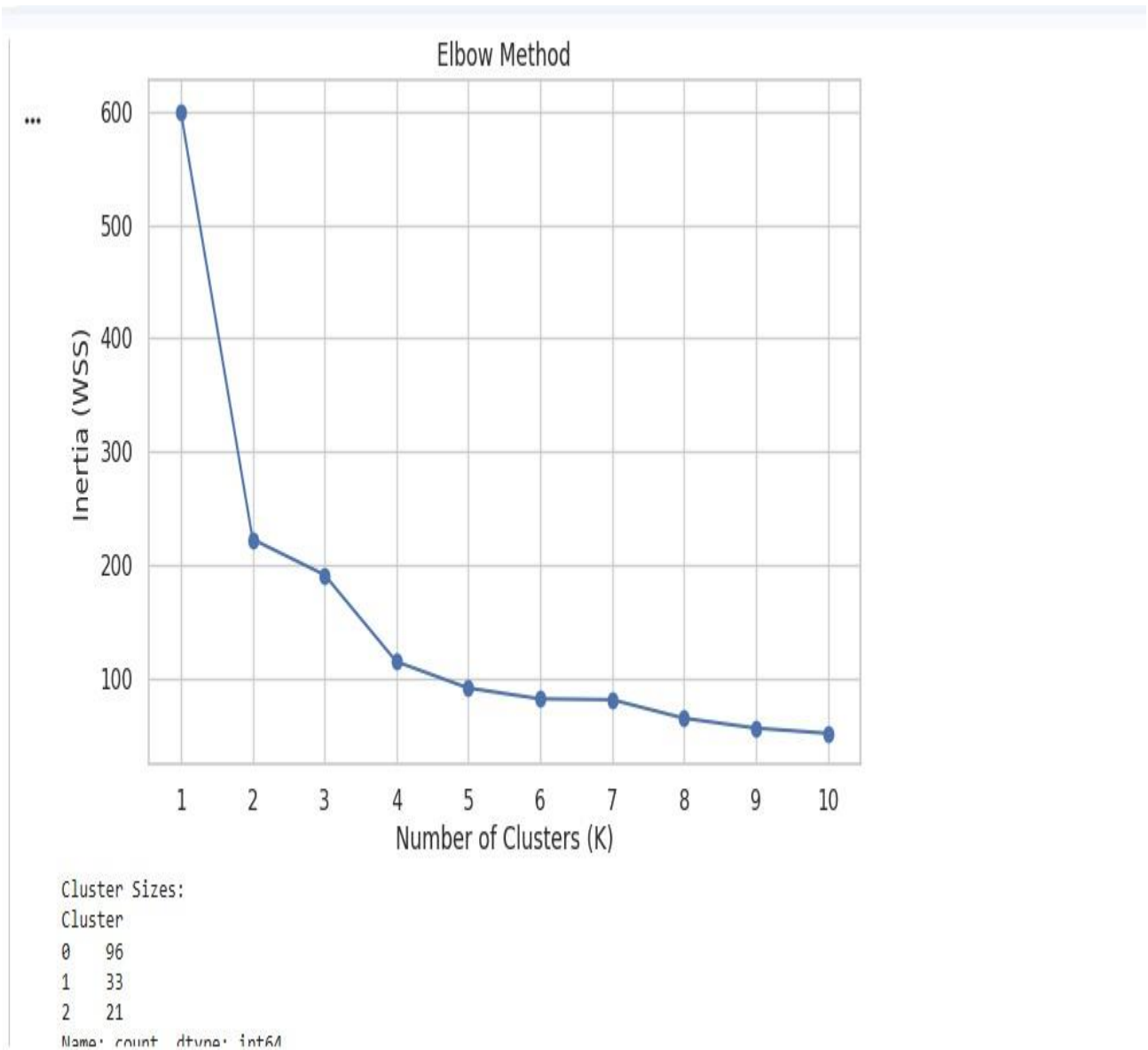
```
Cluster Counts:
```

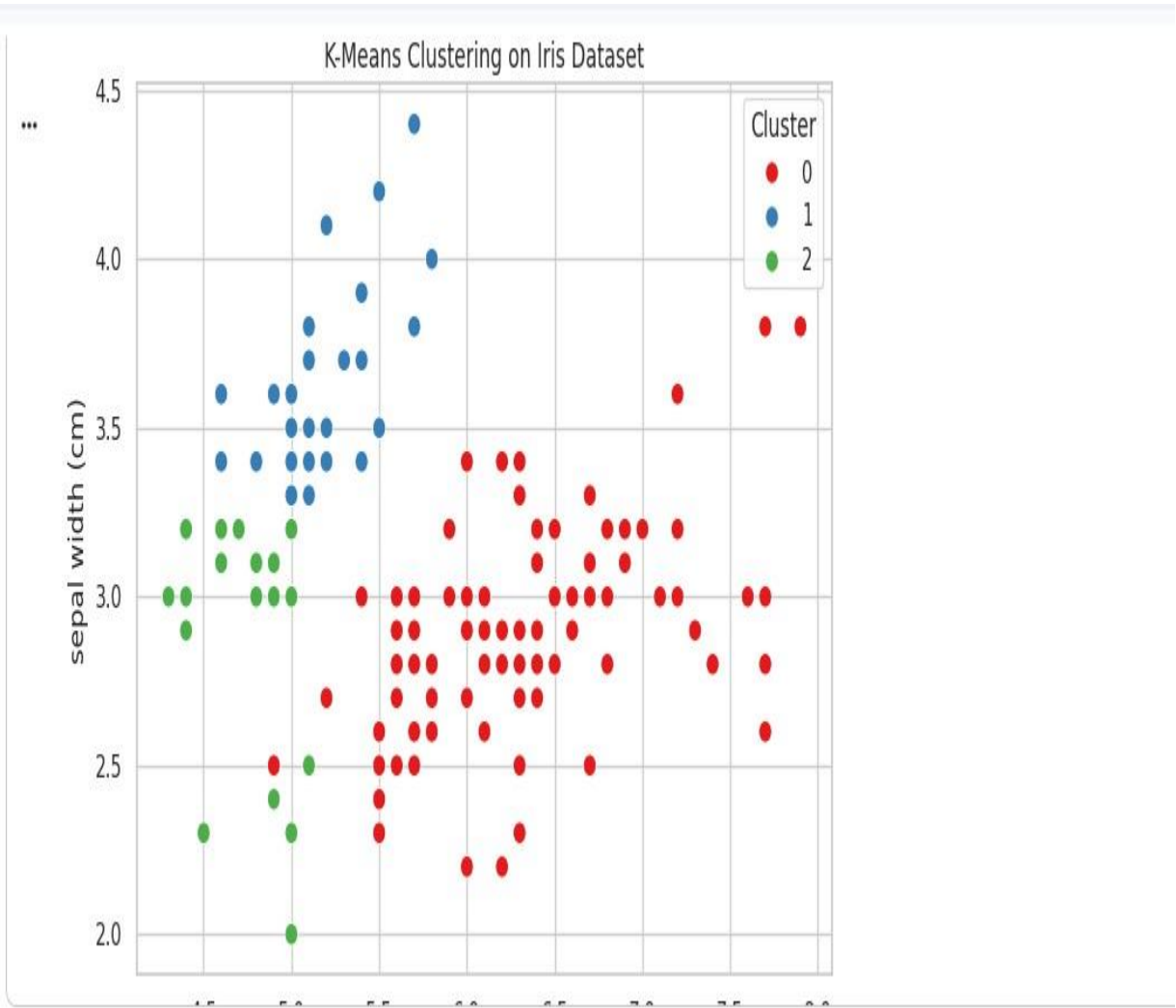
```
Cluster
```

0	96
1	33
2	21

```
Name: count, dtype: int64
```

Q9.





Q10.

```
... First 5 Rows:
   CustomerID  Genre  Age  Annual Income (k$)  Spending Score (1-100)
0           1   Male   19                15                39
1           2   Male   21                15                81
2           3  Female  20                16                 6
3           4  Female  23                16                77
4           5  Female  31                17                40
```

Dataset Shape:
(200, 5)

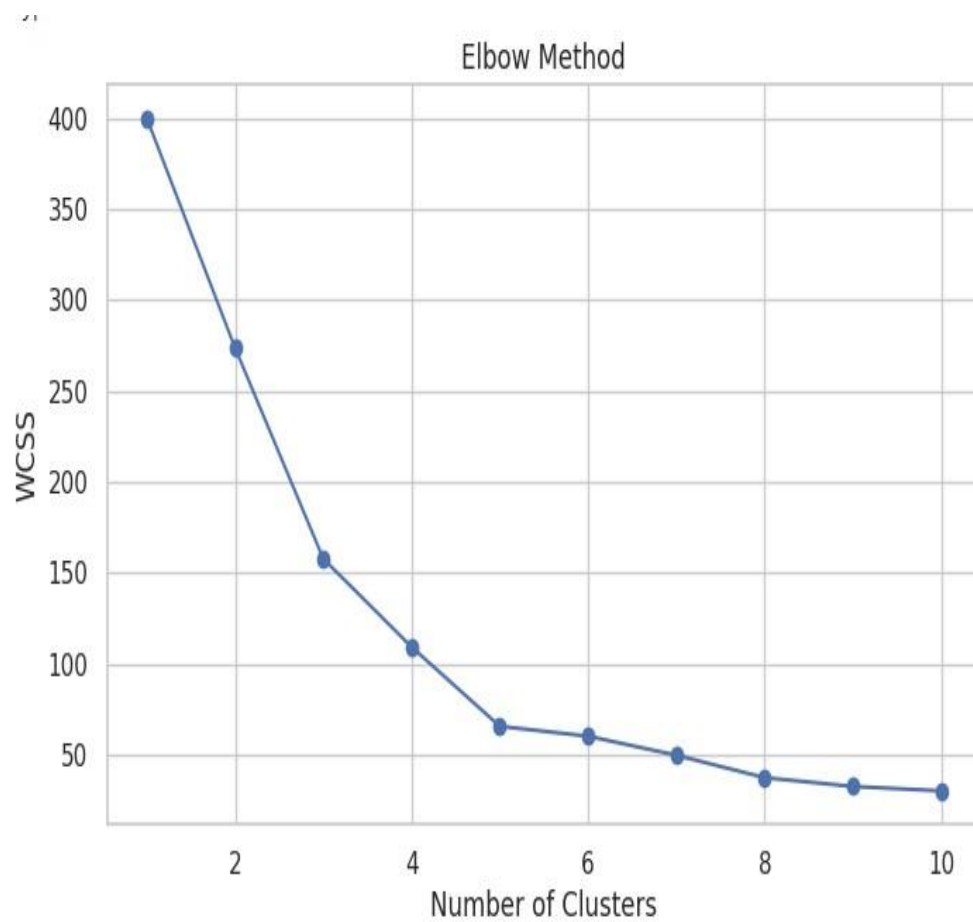
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	CustomerID	200 non-null	int64
1	Genre	200 non-null	object
2	Age	200 non-null	int64
3	Annual Income (k\$)	200 non-null	int64
4	Spending Score (1-100)	200 non-null	int64

dtypes: int64(4), object(1)
memory usage: 7.9+ KB
None

Missing Values:

CustomerID	0
Genre	0
Age	0
Annual Income (k\$)	0
Spending Score (1-100)	0
dtype: int64	



Cluster Sizes:

Cluster

0 81

1 39

2 22

3 35

4 23

Name: count, dtype: int64

\

